ASSIGNMENT – 10 kd2_vaibhav_96218

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle,
Perimeter = 2*(L+W) and Area = L*W.

```
kd2_vaibhav_96218>CREATE procedure sp_area_peri(len INT,wid INT)
    -> BEGIN
    ->  DECLARE prei INT;
    ->  DECLARE area INT;
    ->  SET prei = 2*(len+wid);
    ->  SET area = len*wid;
    ->  SELECT len, wid, prei, area;
    -> END;
    -> $$
ERROR 1304 (42000): PROCEDURE sp_area_peri already exists
kd2_vaibhav_96218>call sp_area_peri(20,14);
    -> ^C
kd2_vaibhav_96218>delimiter ;
kd2_vaibhav_96218>call sp_area_peri(20,14);
+------+------+------+------+
| len  | wid  | prei | area |
+------+------+------+------+
|   20 |   14 |   68 |  280 |
+------+------+------+------+
1 row in set (0.046 sec)

Query OK, 0 rows affected (0.086 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
kd2_vaibhav_96218>DELIMITER $$
kd2_vaibhav_96218>CREATE procedure sp_squ_cube(arg INT)
    -> BEGIN
    ->  DECLARE squ INT;
    ->  DECLARE cub INT;
    ->  SELECT arg*arg INTO squ;
    ->  SELECT arg*arg*arg INTO cub;
    ->  SELECT arg number, squ square, cub cube_NUM;
    -> END;
    -> $$
Query OK, 0 rows affected (0.088 sec)

kd2_vaibhav_96218>DELIMITER ;
kd2_vaibhav_96218> CALL sp_squ_cube(6);
+--------+--------+----------+
| number | square | cube_NUM |
+--------+--------+----------+
|      6 |     36 |      216 |
+--------+--------+----------+
1 row in set (0.053 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are

C= (F-32) *5/9 F= 9/5*C + 32

```
kd2_vaibhav_96218>CREATE PROCEDURE conv_cel_fah(arg float, ch char)
    -> BEGIN
    ->  DECLARE  c float;
    ->  DECLARE f float;
    ->  IF 'f' = ch THEN
    ->          SET C = (arg-32)*5/9 ;
    ->  ELSE
    ->          SET F = 9/5*arg+32 ;
    ->  END IF;
    ->  SELECT arg, ch , c Celsius, f Fahrenheit;
    -> END;
    -> $$
Query OK, 0 rows affected (0.061 sec)

kd2_vaibhav_96218>
kd2_vaibhav_96218>DELIMITER ;
kd2_vaibhav_96218>
kd2_vaibhav_96218>CALL conv_cel_fah(100, 'f');
+------+-----+---------+------------+
| arg  | ch  | Celsius | Fahrenheit |
+------+-----+---------+------------+
|  100 | f   | 37.7778 |       NULL |
+------+-----+---------+------------+
1 row in set (0.050 sec)

Query OK, 0 rows affected (0.091 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
kd2_vaibhav_96218>DELIMITER $$
kd2_vaibhav_96218>CREATE procedure in_yfetin(num INT)
    -> BEGIN
    ->  DECLARE  yard INT;
    ->  DECLARE foot INT;
    ->  DECLARE inch INT;
    ->  DECLARE rem INT;
    ->  SET yard = num/36;
    ->  SET rem = num%36;
    ->  SET foot = rem / 12;
    ->  SET inch = rem %12;
    ->  SELECT yard, foot, inch;
    -> END;
    -> $$
Query OK, 0 rows affected (0.045 sec)

kd2_vaibhav_96218>
kd2_vaibhav_96218>DELIMITER ;
kd2_vaibhav_96218>call in_yfetin(90);
+------+------+------+
| yard | foot | inch |
+------+------+------+
|    3 |    2 |    6 |
+------+------+------+
1 row in set (0.017 sec)

Query OK, 0 rows affected (0.054 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
kd2_vaibhav_96218>DELIMITER $$
kd2_vaibhav_96218>
kd2_vaibhav_96218>CREATE procedure  check_num(num1 int ,num2 int)
    -> BEGIN
    ->  DECLARE  res int;
    ->  DECLARE  ch char;
    ->  SET res = num1 * num2;
    ->  IF(res > 100) THEN
    ->          SET ch = 'Y';
    ->  else
    ->          SET ch = 'N';
    ->  END IF;
    ->  SELECT num1, num2, res, ch;
    -> END;
    -> $$
Query OK, 0 rows affected (0.051 sec)

kd2_vaibhav_96218>
kd2_vaibhav_96218>DELIMITER ;
kd2_vaibhav_96218>
kd2_vaibhav_96218>CALL check_num(10,20);
+------+------+------+------+
| num1 | num2 | res  | ch   |
+------+------+------+------+
|   10 |   20 |  200 | Y    |
+------+------+------+------+
1 row in set (0.027 sec)

Query OK, 0 rows affected (0.089 sec)
```